

Shengdu Chai

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EDUCATION

- Fudan University** Shanghai, China
• *Bachelor of Science with Honors in Physics;*
GPA(Overall): 3.68/4.0 (Rank: 7/131); GPA(Major): 3.87;
Sept. 2019 - June. 2024
- University of Chicago** Chicago, IL, US
• *Non-Degree Visiting Students Program*
Supervisor: Lian-Tao Wang, Professor
July 2022 - Sept. 2022
- Fudan University** Shanghai, China
• *PhD student in Condensed Matter Theory*
Supervisor: Xincheng Xie, Professor
Sept. 2024 - present

PUBLICATIONS

- [1] **Shengdu Chai**, Jiayin Gu, Lian-Tao Wang, “Accommodating the CDF W -boson mass measurement in the beautiful mirror model”, [Physical Review D](#) **107**, 095013 (2023).
- [2] **Shengdu Chai**, Jiayin Gu, Lingfeng Li, “From optimal observables to machine learning: an effective-field-theory analysis of $e^+e^- \rightarrow W+W^-$ at future lepton colliders”, [Journal of High Energy Physics](#) **2024**, 292 (2024).
- [3] **Shengdu Chai**, Chen Lin, Xinyang Dong, Yuqiang Li, Wanli Ouyang, Lei Wang, X. C. Xie, *Revisiting the Broken Symmetry Phase of Solid Hydrogen: A Neural Network Variational Monte Carlo Study*, Dec. 2025.
- [4] Hongfang Liu, Ming Lu, **Shengdu Chai**, Zhi-Qiang Zhang, Hua Jiang, “Measurement of the Chern number for non-Hermitian Chern insulators”, [Physical Review B](#) **109**, 155404 (2024).
- [5] Wanghan Xu, *Probing Scientific General Intelligence of LLMs with Scientist-Aligned Workflows*, Dec. 2025.

RESEARCH EXPERIENCE

- NNVMC for Beyond-BO Simulations of Hydrogen Solids** Fudan University & Shanghai AI Lab
• *Advised by Assoc. Prof. Chen Lin, Prof. Lei Wang and Prof. Xincheng Xie* May 2024 – Present
 - Developed a general-purpose Neural Network Variational Monte Carlo (NNVMC) framework for high-precision ground-state calculations of hydrogen solids, extending beyond the Born–Oppenheimer (BO) approximation by incorporating Coulomb interactions and nuclear quantum effects.
 - Extended NNVMC to the NPT ensemble, enabling joint optimization of crystal structures and wave functions; scaled simulations from 32 to 200 particles via forward-only geometry optimization.
 - Predicted a candidate crystal structure for hydrogen phase II through NPT ensemble simulations, suggesting the possible existence of a new quantum phase.
- Machine Learning in SMEFT** Fudan University
• *Research Assistant to Associate Professor Jiayin Gu* Jun. 2022 – Jan. 2024
 - Aimed to apply machine learning techniques to the phenomenological analyses of the Standard Model Effective Field Theory (SMEFT), focusing on the measurements at future lepton colliders
 - Performed machine learning methods with simulations of $e^+e^- \rightarrow W^+W^-$, including some systematic effects to determine the likelihood ratio in terms of the Wilson coefficients of dimension-six operators in this process.
 - Determined that the machine learning method performed better than the traditional methods, such as Optimal Observable, which corrected the large bias of model parameters and gave strong constraints
 - Found the bias has a scaling law with the number of training samples, with more training samples, the bias can be reduced to a negligible level
- Explanation of New CDF W Mass** University of Chicago
• *Research Assistant to Professor Lian-Tao Wang* July 2022 – Sept. 2022
 - Explained both the new W boson mass $m_W^{\text{CDF-II}}$ reported by Fermi Lab and the long existed discrepancy of forward-backward asymmetry $A_{FB}^{0,b}$ by introducing new vector-like quarks
 - Explored the model properties by performing a global electroweak fit. Found that the model is consistent with the current direct-search limits at the LHC, the HL-LHC, can cover most of the regions of the parameter space preferred by the electroweak fit. Determined that the one-loop contribution to Higgs couplings in this model was also relevant, which is consistent on current measurement and may be excluded on future collider
 - Determined that the mass of the exotic quark (with charge $-4/3$) is required to be below 4 TeV at the 95% confidence level, and the best-fit point corresponded to a mass of around 1.5 TeV

TALKS AND SEMINARS

- **Invited Talks**

- Probing BSM effects in $e^+e^- \rightarrow WW$ with machine learning
* HKUST ias *Oct.2022*

- **Contributed Talks**

- **IHEP** Workshop on the High Energy Circular Electron Positron Collider
<https://indico.ihep.ac.cn/event/17020/contributions/119266/> *Oct.2022*
- **HKUST ias** IAS Program on High Energy Physics
<https://indico.cern.ch/event/1215937/timetable/#20230212.detailed> *Feb.2023*
- **IAIFI** IAIFI Summer School Lightning Talk *Aug.2024*

- **Posters**

- **BIMSA** The first International Congress of Basic Science (ICBS)
<https://www.icbs.cn/en/web/index/18009> *July.2023*
- **USTC** The eleventh PFUNT Conference *Aug.2024*

HONORS AND AWARDS

- **National Scholarship**(top 10%) *Oct.2024*
- **Outstanding Graduate of Fudan University**(top 10%) *Jun.2024*
- **Bachelor of Science with Honors**(top 5%) *Jun.2024*
- **Fudan University Undergraduate Professional Scholarship** *Oct.2022*
- **Honors Student in Department of Physics** *Oct.2022*
- **Fudan University Undergraduate Professional Scholarship** *Oct.2021*
- **Honors Student in Department of Physics** *Oct.2021*
- **1st Prize in Fudan University Scholarship for Outstanding Students**(top 4 in the Department) *Sept.2021*
- **Outstanding Student in Fudan University** *Oct.2020*
- **2nd Prize of China Undergraduate Physics Tournament**,a team-oriented physics competition between 60 top universities in China *Oct.2020*
- **1st Prize in Fudan University Scholarship for Outstanding Students**(top 5 in the Department) *Sept.2020*
- **2nd Prize of 2020 Mathematical Contest in Modeling** ,
a team-oriented competition of math modeling *Sept.2020*
- **Outstanding Student in Fudan University** *May.2020*

SKILLS SUMMARY

- **Programming Skills:** Python (proficient), C/C++, pytorch, Mathematica, jax
- **Computer Skills:** Latex, Machine Learning, COMSOL, Root, MadGraph 5, Delphes